

ECO 202 — Practice Exam 3 — Version A

Statistics and Data Analysis for Economics | Princeton University

Fall 2026 | Classes 10–13 | 100 points

Name: _____

Student ID: _____

Timing. Target workload: 70 minutes (10 minutes multiple choice; 60 minutes written work). Follow announced start/end times within the 80-minute meeting. For practice, set a 70-minute timer.

Conditions. No books, notes, calculators, AI, or other electronics; University-authorized accommodations apply. Fractions, radicals, and equivalent exact expressions are acceptable. Show reasoning in written answers. Data are teaching examples unless identified otherwise.

Points. Questions 1–4: one answer A–D, 5 points each, no negative marking. Questions 5–8: 20 points each; subpart points are shown. Label any continuation clearly.

Multiple choice (20 points; about 10 minutes)

1. Let $H = \mathbf{1}\{X > 4\}$. Which identity always holds?
 - A. $\mathbb{E}[H] = \mathbb{E}[X]$.
 - B. $\mathbb{E}[H]$ must be either zero or one.
 - C. $\mathbb{E}[H] = \mathbb{P}(X > 4)$.
 - D. $\text{Var}(H) = \mathbb{P}(X > 4)$.
2. An independent sample of 64 observations has sample standard deviation 12. What is the usual estimated standard error of its sample mean?
 - A. 12.
 - B. $3/2$.
 - C. $12/64$.
 - D. $144/64$.
3. Independent, identically distributed observations have a skewed distribution with finite positive variance. Which is the appropriate CLT statement?
 - A. For every $n > 30$, the sample mean is exactly Normal.
 - B. The individual observations become approximately Normal as n grows.
 - C. The unstandardized sample mean approaches a standard Normal distribution.
 - D. The centered sample mean divided by its true standard error approaches a standard Normal distribution.
4. Two variables with finite variances have zero covariance. What follows?
 - A. They must be independent.
 - B. One of them must be constant.
 - C. Their joint distribution is determined by their means and variances.
 - D. The variance of their sum equals the sum of their variances.

5. Marginal distributions and transformations (20 points; about 15 minutes)

The following table gives joint probabilities.

	$X = 0$	$X = 2$
$Y = 0$	$1/4$	$1/4$
$Y = 2$	$1/8$	$1/8$
$Y = 4$	$1/4$	0

(a) (6 points) Find the marginal distribution of Y , its expectation, and its variance.

(b) (4 points) Find the full distribution of $V = (Y - 2)^2$. Why can different values of Y contribute to the same value of V ?

- (c) **(6 points)** Calculate $\mathbb{E}[X]$, $\mathbb{E}[XY]$, and $\text{Cov}(X, Y)$. Explain the sign of the covariance in this table.
- (d) **(4 points)** You may use $\text{Var}(X) = 15/16$. Find $\text{Var}(Y - X)$ and explain why subtracting X increases variability here relative to adding it.

6. From two draws to many counts (20 points; about 15 minutes)

A teaching population has two values, 0 and 2, each selected with probability $1/2$. Draw independently with replacement. Write X_i for draw i and $H_i = \mathbf{1}\{X_i = 2\}$.

Use $\Phi(z) = \mathbb{P}(Z \leq z)$ for standard Normal Z , with $\Phi(9/8) \approx 0.8697$ and $\Phi(1) \approx 0.8413$.

(a) (6 points) List all four possible ordered samples of size two and obtain the sampling distribution of $\bar{X}_2$. Calculate its expectation and variance directly from this distribution.

(b) (6 points) Now take 64 independent draws. For $K = \sum_{i=1}^{64} H_i$ and $\hat{P} = K/64$, calculate the means and standard deviations of K and $\hat{P}$. State the exact distribution family of K and its parameters.

- (c) **(8 points)** Approximate $\mathbb{P}(28 \leq K \leq 36)$ using a continuity correction at *both* boundaries. Show the standardized interval and probability. Explain what the correction accomplishes and one reason the result remains approximate.

7. What averaging does and does not do (20 points; about 15 minutes)

Customer expenditure amounts X_i are independent and identically distributed, with population mean 20 dollars and standard deviation 12 dollars. Their distribution is right-skewed. Let $\bar{X}_n$ be the mean of n observations.

Use $\Phi(1) \approx 0.8413$ and $\Phi(2) \approx 0.9772$ for the standard Normal CDF.

(a) (6 points) Derive $\mathbb{E}[\bar{X}_{36}]$, $\text{Var}(\bar{X}_{36})$, and $\text{SE}(\bar{X}_{36})$. State which results are exact under the model and the units of the standard error.

(b) (6 points) Use the CLT to approximate $\mathbb{P}(\bar{X}_n > 22)$ for $n = 36$ and for $n = 144$. Interpret the change while keeping the threshold and population fixed.

- (c) (4 points) A student says, “The LLN makes the original expenditure distribution less skewed when we collect more observations.” Correct the statement and state what the LLN actually says becomes concentrated.
- (d) (4 points) Suppose observations instead come in groups whose members tend to spend above or below the mean together. Explain which step in the simple standard-error derivation requires re-examination. Does a large row count alone justify using $12/\sqrt{n}$?

8. Bias, precision, and the target (20 points; about 15 minutes)

A model treats 100 responses to a renewal-*intention* question as independent Bernoulli draws with common yes probability p . The observed data have 64 yes responses. Let $\hat{P}$ be the sample proportion and consider $T = (\hat{P} + 1/2)/2$ as another estimator of p .

You may use $\text{MSE}(T; p) = \text{Var}(T) + (\mathbb{E}[T] - p)^2$ and the estimated standard error $\sqrt{\hat{p}(1 - \hat{p})/100}$ for the sample proportion.

- (a) **(5 points)** Identify the estimand, estimator $\hat{P}$, and realized estimate. Calculate the estimated standard error of $\hat{P}$ from the observed sample.
- (b) **(5 points)** Derive the bias and variance of T as functions of p . Compare its variance with that of $\hat{P}$.

- (c) **(5 points)** Compare the MSEs of T and $\hat{P}$ at $p = 1/2$ and at $p = 0$. These are hypothetical model values for evaluating the rules; they are not claims about the observed sample. Does either unbiasedness alone or variance alone settle the comparison?
- (d) **(5 points)** Explain why an unbiased estimator of p does not necessarily estimate the proportion who *actually renew later* without bias. Would increasing the number of intention responses resolve that distinction?